Installing and updating the VPN client

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Summary

This article guides users through installing, updating, and troubleshooting the Palo Alto GlobalProtect VPN client on Acme Global managed devices. It covers common symptoms such as connection failures, frequent drops, authentication errors, certificate expiry, and inability to reach internal resources, and provides self-help steps to resolve most issues. Service-desk agents can follow a dedicated procedure to verify certificates, reinstall the client, and escalate when needed.

Symptoms

- VPN client shows "Connection failed" or "Unable to connect to vpn.acme.example"
- Frequent disconnections after a few minutes of use
- Authentication failed prompts despite correct credentials
- Certificate expired warning in the client UI
- Connected to VPN but cannot ping or browse internal sites (intranet, file shares)

Common causes

- Expired or corrupted machine certificate used for GlobalProtect authentication
- Outdated VPN client version lacking recent security patches
- Misconfigured split-tunnel or DNS settings causing traffic to bypass the tunnel
- Local network restrictions (home router firewall, ISP blocking UDP 4500/500 or TCP 443)
- Corrupted client profile or cached credentials

Pre-checks

Before proceeding with installation or troubleshooting, verify that the device meets the basic requirements and that you have the necessary permissions. Ensure you are logged in with your Acme Global corporate account and that the device is enrolled in Intune. Confirm that you have internet access (e.g., you can browse public websites) and that the GlobalProtect portal address vpn.acme.example is reachable (try pinging it or opening https://vpn.acme.example in a browser).

Self-Help Procedure: Install or Update GlobalProtect

Follow these steps to install the VPN client for the first time or to update to the latest version available through the Company Portal. The procedure works on both Windows and macOS devices managed by Intune. If you encounter any error messages during the process, note them and proceed to the Service Desk section.

1. Open the Company Portal app (Windows: Start -> Company Portal; macOS: Applications -> Company Portal).
2. Search for "GlobalProtect" in the app catalog.
3. Select the GlobalProtect entry and click "Install". The portal will download the appropriate installer for your OS.
4. Once the download completes, the installer will launch automatically. Accept the license agreement and keep the default installation folder.
5. After installation finishes, launch GlobalProtect from the Start menu (Windows) or Applications folder (macOS).
6. When prompted, enter the portal address: vpn.acme.example and click "Connect".
7. If a certificate prompt appears, choose the machine certificate issued by Acme Global (usually listed as "Acme Global Machine Certificate").
8. The client should establish a connection and display a green shield icon. Verify connectivity by accessing an internal site such as https://intranet.acme.example.

Service Desk Procedure: VPN Troubleshooting

Use this procedure when a user reports persistent VPN issues after attempting the self-help steps. It guides the agent through certificate validation, client reinstallation, DNS flush, and split-tunnel verification. Document each step in the

ticket and collect logs before escalating to Network Support if the problem persists.

1. Ask the user to reproduce the issue and capture any exact error messages (e.g., "Authentication failed", "Certificate expired").
2. Verify the user's device is compliant in Intune (check device compliance policy).
3. Check the GlobalProtect client version: open GlobalProtect -> Settings -> About. Compare against the latest version published in the Company Portal.
4. If the version is outdated, initiate a reinstall via Company Portal as described in the self-help procedure.
5. Inspect the machine certificate: run certmgr.msc (Windows) or Keychain Access (macOS) -> Certificates -> Personal. Ensure the certificate issued to the device is valid and not expired; note the expiry date.
6. If the certificate is expired or missing, request a new certificate from the Identity & Access Management team using the Temporary Access Pass process.
7. Flush DNS resolver cache: Windows - ipconfig /flushdns; macOS - sudo killall -HUP mDNSResponder.
8. Verify split-tunnel configuration: In GlobalProtect settings, confirm that "Split Tunnel" is enabled and that the correct network routes are listed (typically 10.0.0.0/8 for internal Acme networks).
9. Test connectivity: ping an internal server (e.g., fileserver.acme.example) and attempt to access an internal web application.
10. If the connection still fails, collect GlobalProtect logs (Settings -> Diagnostics -> Export Log) and attach them to the ticket.

Verification

After completing either the self-help or service-desk steps, confirm that the VPN is functioning correctly and that the user can resume normal work. This verification helps prevent repeat tickets and ensures that the underlying issue has been resolved.

1. Have the user connect to GlobalProtect and observe the connection status indicator (green shield).
2. Attempt to access at least two distinct internal resources (e.g., intranet homepage and a file share).
3. Ask the user to perform a typical work task that requires VPN, such as opening a remote desktop session to a lab machine or accessing a protected SharePoint site.
4. If all tests succeed, close the ticket with a resolution note summarizing the actions taken (certificate renewed, client updated, DNS flushed, etc.).

When to escalate

Escalate to the Network Support team if the VPN connection remains unsuccessful after completing the Service Desk Procedure, or if the user experiences repeated certificate expirations despite renewal. Include in the escalation: the exact error messages, GlobalProtect client version, certificate validity dates, Intune compliance status, and the exported GlobalProtect logs. Provide the user's name, employee ID, device hostname, and the ticket number. Network Support will investigate potential infrastructure issues such as GlobalProtect gateway health, certificate authority problems, or network-level blocks.

Keywords: VPN, GlobalProtect, Palo Alto, install VPN, update VPN, certificate expired, authentication failed, VPN disconnects, split tunnel, DNS flush, Intune, Company Portal, Network Support, vpn.acme.example